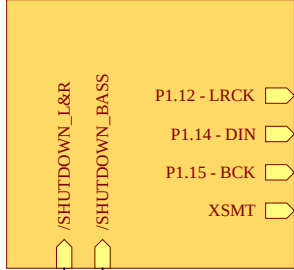
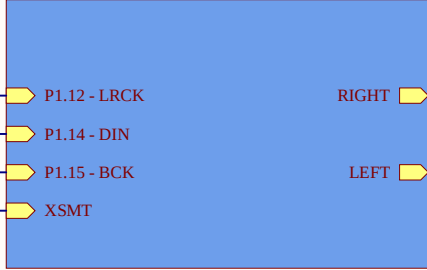


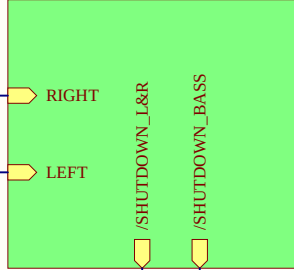
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nRF5340.SchDoc



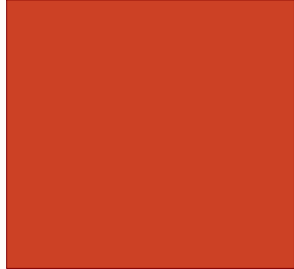
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DAC.SchDoc



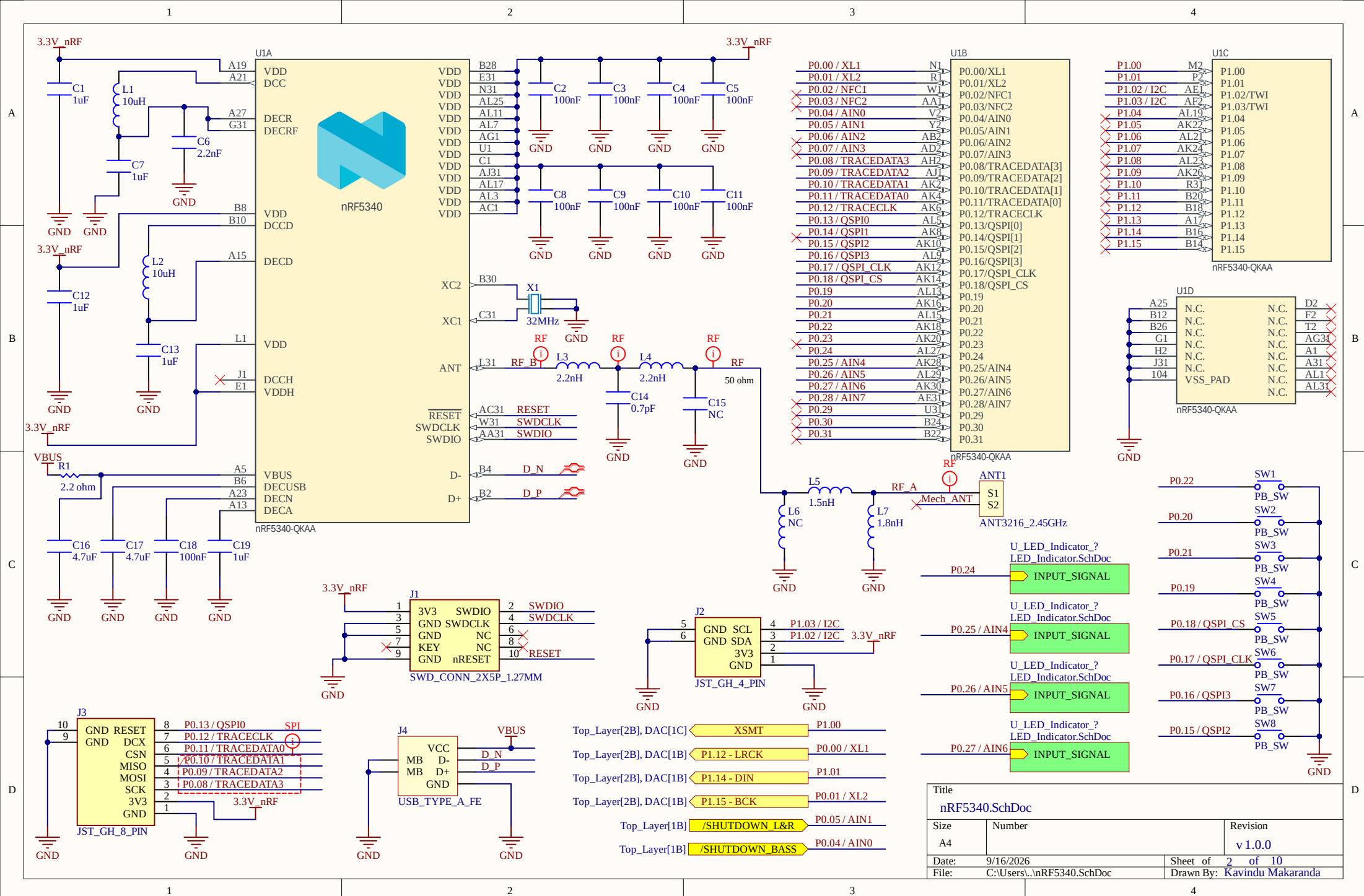
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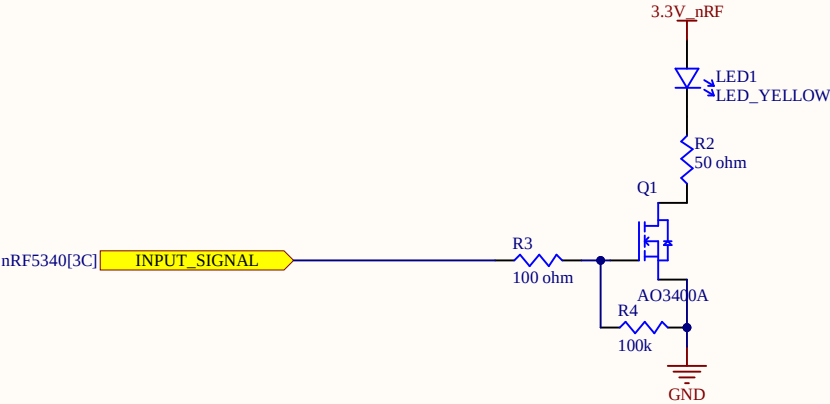


U_Power_Section
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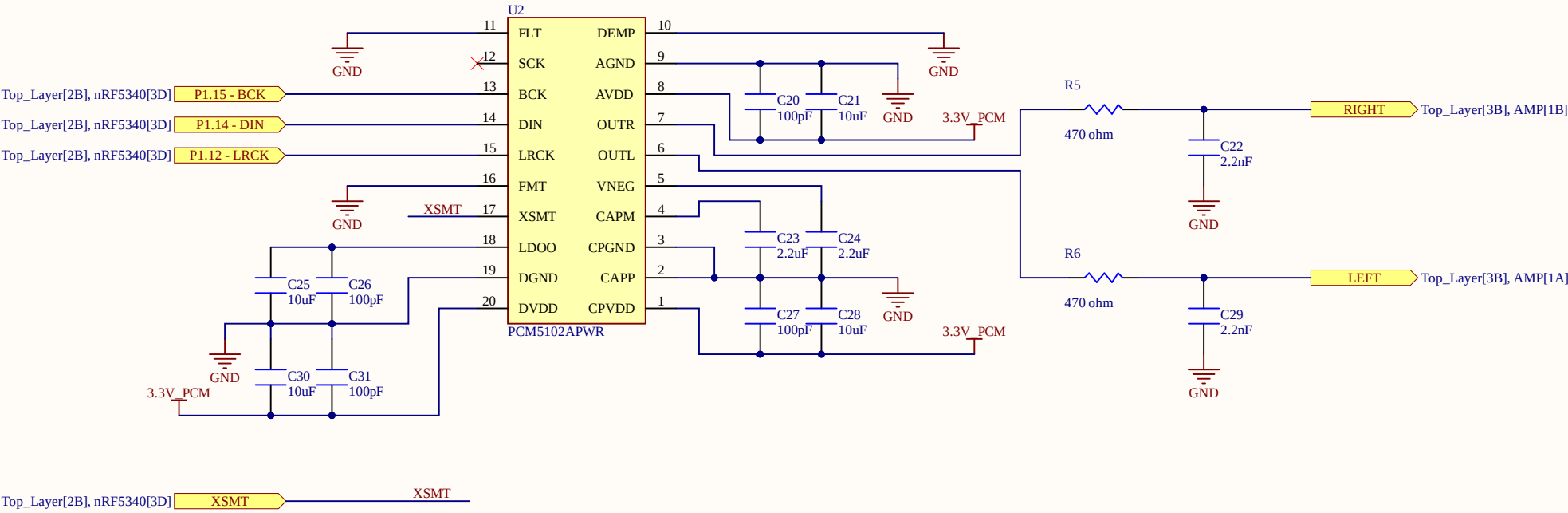
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Size A4	Number		Revision v 1.0.0
Date:	9/16/2026	Sheet of	1 of 10
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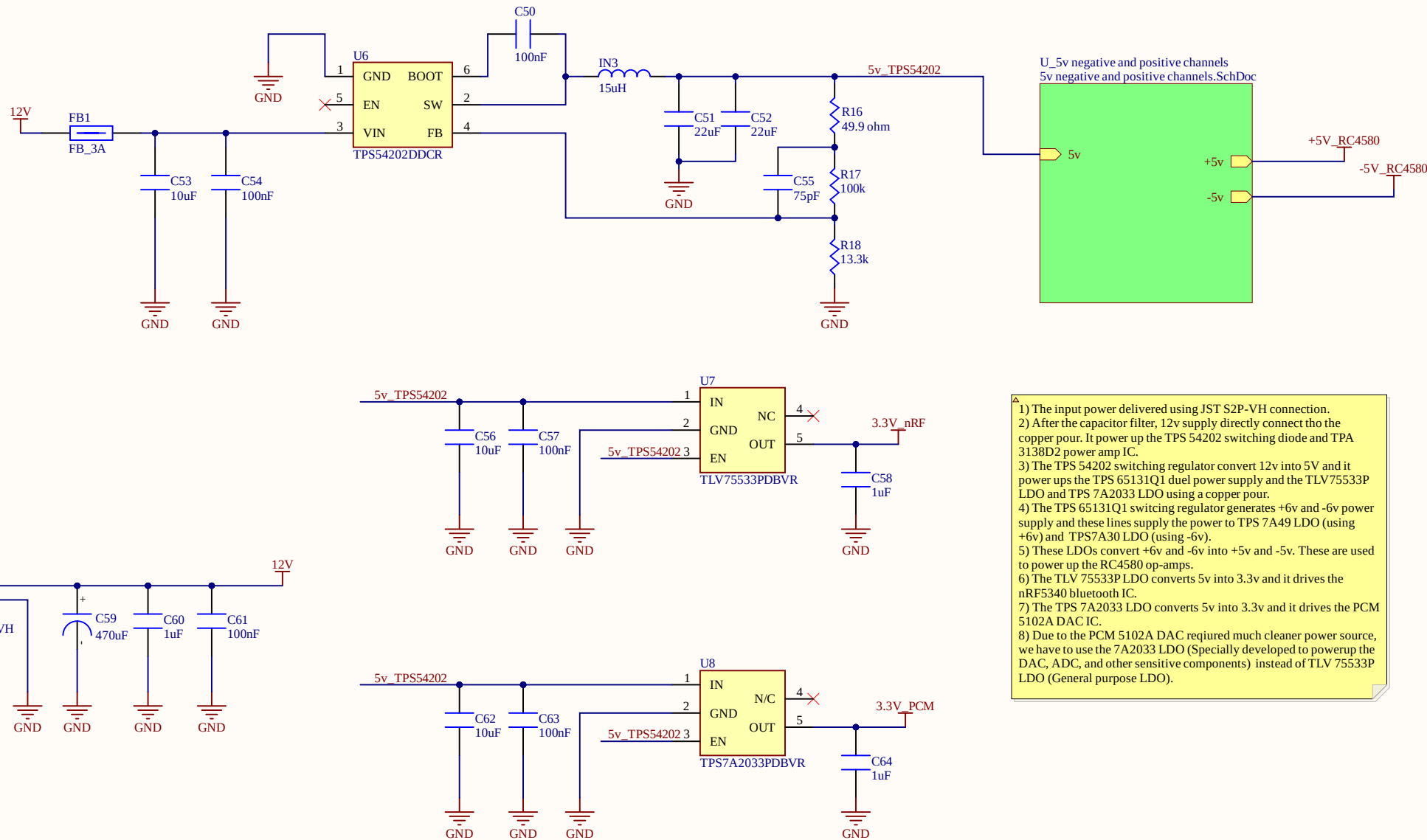


Title			
LED_Indicator.SchDoc			
Size	Number		Revision
A4			v 1.0.0
Date:	9/16/2026	Sheet of	3 of 10
File:	C:\Users\...\LED_Indicator.SchDoc	Drawn By:	Kavindu Makaranda

For XSMT pin,
If the pin is high (3.3v), the system works fine and stays at unmuted state.
If the XSMT pin makes a transition from "1" to "0" over 6 ms or more, the device switches into external under voltage protection mode. This mode uses two trigger levels,
i) XSMT pin level reaches 2 V, soft mute process begins.
ii) XSMT pin level reaches 1.2 V, analog mute engages, regardless of digital audio level, and analog shutdown begins. (DAC and related circuitry powers down).
If XSMT is moved from "1" to "0" in 20 ns or less, then the device will interpret it as a digital controlled request to mute. It will perform a soft mute, then move to standby.

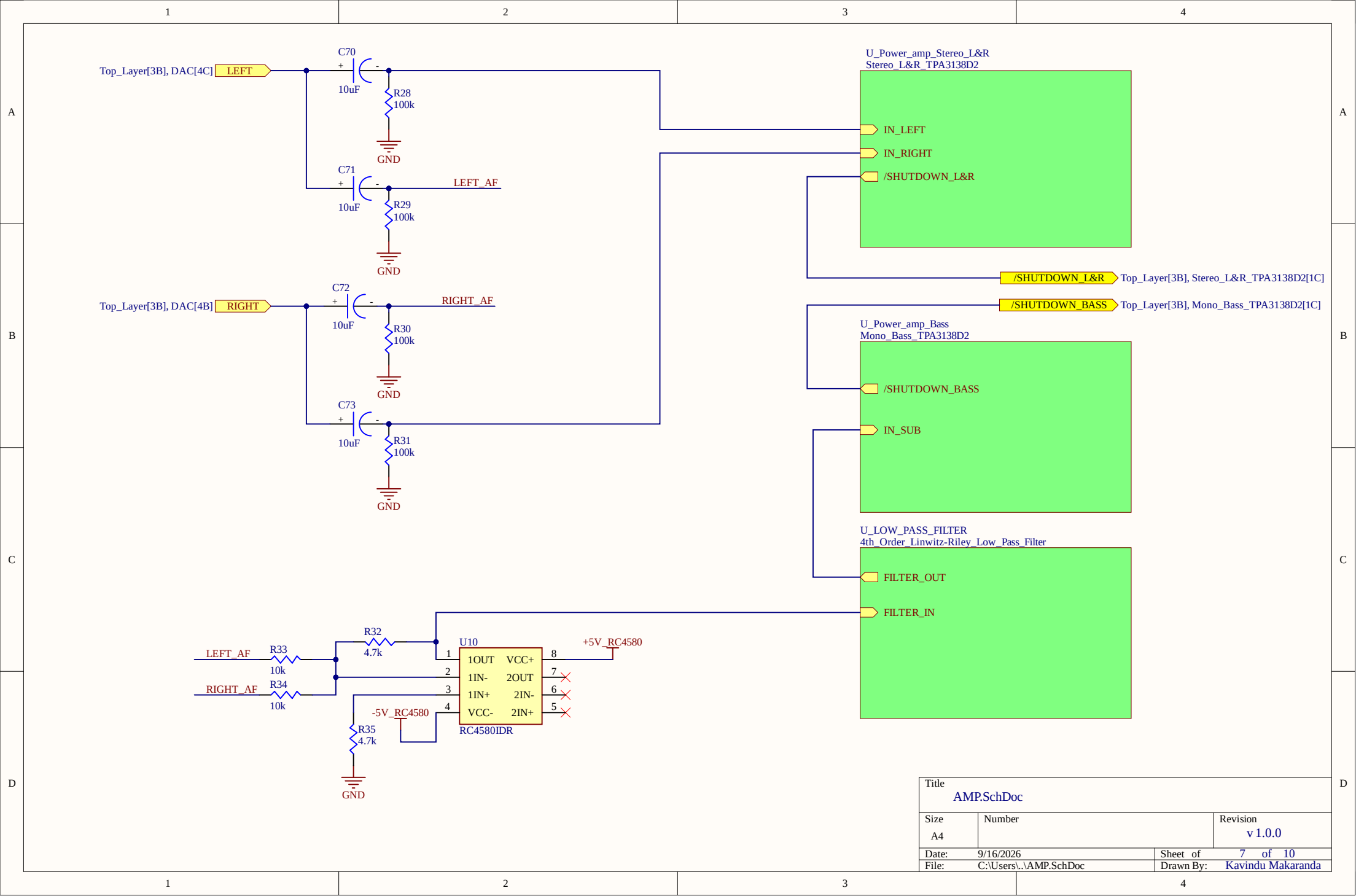


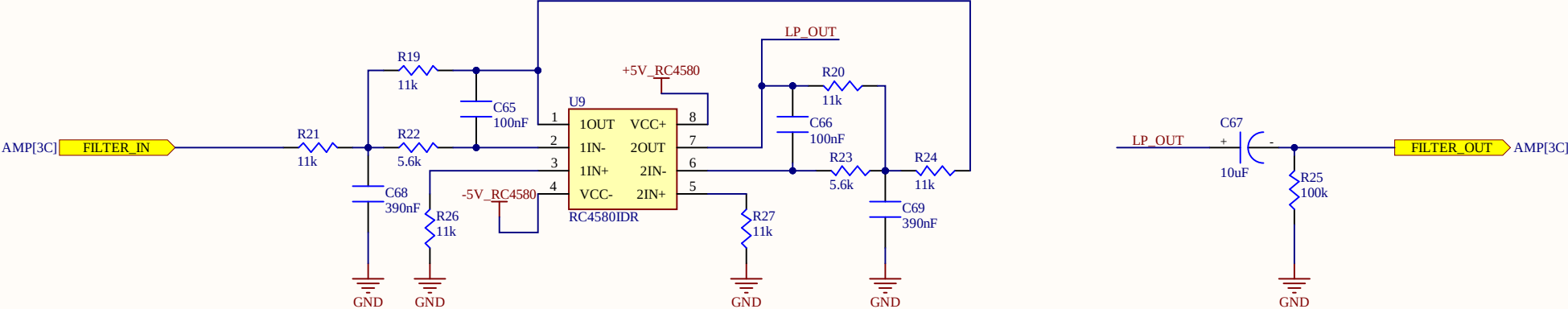
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Size A4	Number	Revision v 1.0.0
Date: 9/16/2026	Sheet of 4 of 10	
File: C:\Users\...\DAC.SchDoc	Drawn By: Kavindu Makaranda	



- 1) The input power delivered using JST S2P-VH connection.
- 2) After the capacitor filter, 12v supply directly connect tho the copper pour. It power up the TPS 54202 switching diode and TPA 3138D2 power amp IC.
- 3) The TPS 54202 switching regulator convert 12v into 5V and it power ups the TPS 65131Q1 duel power supply and the TLV75533P LDO and TPS 7A2033 LDO using a copper pour.
- 4) The TPS 65131Q1 switching regulator generates +6v and -6v power supply and these lines supply the power to TPS 7A49 LDO (using +6v) and TPS7A30 LDO (using -6v).
- 5) These LDOs convert +6v and -6v into +5v and -5v. These are used to power up the RC4580 op-amps.
- 6) The TLV 75533P LDO converts 5v into 3.3v and it drives the nRF5340 bluetooth IC.
- 7) The TPS 7A2033 LDO converts 5v into 3.3v and it drives the PCM 5102A DAC IC.
- 8) Due to the PCM 5102A DAC required much cleaner power source, we have to use the 7A2033 LDO (Specially developed to powerup the DAC, ADC, and other sensitive components) instead of TLV 75533P LDO (General purpose LDO).

Title Power_Section.SchDoc		
Size A4	Number	Revision v 1.0.0
Date: 9/16/2026	Sheet of 5 of 10	
File: C:\Users\...\Power_Section.SchDoc	Drawn By: Kavindu Makaranda	





Title		
4th_Order_Linwitz-Riley_Low_Pass_Filter.SchDoc		
Size	Number	Revision
A4		v 1.0.0
Date:	9/16/2026	Sheet of 8 of 10
File:	C:\Users\...4th_Order_Linwitz-Riley_Low_Pass_Filter.SchDoc	
Bhavindu Makaranda		

